FROM ALMOST COMPLEX STRUCTURES TO COMPLEX MANIFOLDS

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Introduction

In the first lecture of the semester we learned a natural way to introduce the complex numbers: equip the set $\mathbb{R}^2$ with the multiplication

$$(x, y) \cdot (u, v) = (xu - yv,\ xv + yu),$$

and identify (x, y) with $x + iy$, where $i^2 = -1$. In this viewpoint $i = (0, 1) \in \mathbb{R}^2$, and

$$i^2 = (0, 1) \cdot (0, 1) = (-1, 0),$$

which is naturally identified with -1.

However, one sort of notation you might see in many places "iv" for a tangent vector $v \in T_p\mathbb{R}^2$ is *not* a priori well-defined: the symbol i lives on the base while v lives in a tangent fibre. Though it is true that we could regard $T_p\mathbb{R}^2$ as $\mathbb{R}^2$ but this is definitely not natural in geometry, to understand this in the essence, we firstly explain how to regard $\mathbb{C}$ as a Lie group.

$\mathbb{C}$ is *not* a Lie group under multiplication, since $(0, 0)$ has no multiplicative inverse. Hence we should not use multiplicative left translations to build fibrewise maps on $T\mathbb{C}$. When we regard $\mathbb{C}$ as a Lie group, we always mean the additive group $(\mathbb{C}, +)$.

However, the "multiply by i" map is still useful as now

$$\phi :\ (\mathbb{C}, +) \longrightarrow (\mathbb{C}, +), \qquad \phi(z) = i\, z,$$

is a group homomorphism:

$$\phi(x + y) = i(x + y) = ix + iy, \qquad \phi(0) = 0.$$

Therefore at the identity $e = 0$ the differential

$$A := d\phi_e :\ T_e\mathbb{C} \longrightarrow T_e\mathbb{C}$$

is a linear automorphism (in the standard basis it is the 90° rotation, so $A^2 = -\mathrm{Id}$).

Let L_p denote additive left translation $L_p(q) = p+q$ (so dL_p identifies $T_0\mathbb{C}$ with $T_p\mathbb{C}$). We obtain a natural fibrewise endomorphism by the conjugation formula

$$J_p \;:=\; dL_p \,\circ\, A \,\circ\, dL_{-p} \;:\; T_p\mathbb{C} \longrightarrow T_p\mathbb{C}, \tag{0.1}$$

which assembles to a smooth bundle endomorphism $J : T\mathbb{C} \to T\mathbb{C}$. Since $dL_p = \mathrm{Id}$ for translations, (0.1) gives $J_p \equiv A$ (constant in p), and in particular $J^2 = -\mathrm{Id}$.

With this J in hand, the scalar action of a complex number on a tangent vector is well-defined: for $v \in T_p\mathbb{C}$ and $a, b \in \mathbb{R}$,

$$(a + ib) \cdot v \;:=\; a\, v \;+\; b\, J_p v. \tag{0.2}$$

Remark 0.3. For a smooth map $f : M \to M$, the induced map on the tangent bundle

$$Tf : TM \to TM, \qquad (p, v) \mapsto \big(f(p),\, df_p(v)\big)$$

does *not* map each fibre to itself. To obtain a global fibrewise endomorphism one needs the Lie group structure: left translations identify fibres, and an endomorphism $A : \mathfrak{g} \to \mathfrak{g}$ at the identity extends to

$$J_p \;=\; dL_p\, A\, dL_{p^{-1}} :\; T_pG \to T_pG.$$

The bundle endomorphism constructed above extends naturally to the general setting and leads to the standard notion of an almost complex structure:

Definition 0.4 (Almost complex structure)**.** Let M be a smooth manifold. An *almost complex structure* on M is a smooth bundle endomorphism

$$J : TM \to TM$$

such that for every $p \in M$, the linear map $J_p : T_pM \to T_pM$ satisfies

$$J_p^2 = -\mathrm{Id}_{T_pM}.$$

Definition 0.5 (Almost complex manifold)**.** A smooth manifold M together with an almost complex structure J is called an *almost complex manifold* and is denoted (M, J).

1. Complexification and (p, q)-types

Given an almost complex manifold (M, J), we complexify the tangent bundle:

$$T^{\mathbb{C}}M := TM \otimes_{\mathbb{R}} \mathbb{C}.$$

The map $J : TM \to TM$ extends $\mathbb{C}$-linearly to

$$J^{\mathbb{C}} : T^{\mathbb{C}}M \to T^{\mathbb{C}}M, \qquad J^{\mathbb{C}}(v \otimes z) := (Jv) \otimes z.$$

Since $J^2 = -\mathrm{Id}$, we have

$$\big(J^{\mathbb{C}}\big)^2 (v \otimes z) = J^{\mathbb{C}}\big((Jv) \otimes z\big) = J^2 v \otimes z = -v \otimes z,$$

so $(J^{\mathbb{C}})^2 = -\mathrm{Id}$ on $T^{\mathbb{C}}M$.

We make a notational remark. After complexification, any element of $T^{\mathbb{C}}_x M$ can be written as

$$u + iv \quad \text{with } u, v \in T_xM,$$

where this really means

$$u := u \otimes 1, \qquad iv := v \otimes i.$$

So i acts on the *scalar factor* in $T_xM \otimes_{\mathbb{R}} \mathbb{C}$, not directly on $v \in T_xM$.

With this convention, for $u, v \in T_xM$ we may write

$$J^{\mathbb{C}}(u + iv) = J^{\mathbb{C}}(u \otimes 1) + J^{\mathbb{C}}(v \otimes i) = Ju \otimes 1 + Jv \otimes i = Ju + iJv.$$

Thus $J^{\mathbb{C}}$ acts $\mathbb{C}$-linearly on $T^{\mathbb{C}}M$.

Since $J_x^{\mathbb{C}} : T_x^{\mathbb{C}}M \to T_x^{\mathbb{C}}M$ is $\mathbb{C}$-linear and satisfies $(J_x^{\mathbb{C}})^2 = -\mathrm{Id}$, its eigenvalues are $\pm i$. We therefore define

$$T_x^{1,0}M := \ker\big(J_x^{\mathbb{C}} - i\,\mathrm{Id}\big), \qquad T_x^{0,1}M := \ker\big(J_x^{\mathbb{C}} + i\,\mathrm{Id}\big),$$

and we have the direct sum decomposition

$$T_x^{\mathbb{C}}M \;=\; T_x^{1,0}M \;\oplus\; T_x^{0,1}M.$$

Proposition 1.1. *For any real tangent vector $X \in T_xM$, we have*

$$X - i\,JX \in T_x^{1,0}M, \qquad X + i\,JX \in T_x^{0,1}M.$$

Equivalently,

$$T_x^{1,0}M = \{\, X - iJX \mid X \in T_xM \,\}, \qquad T_x^{0,1}M = \{\, X + iJX \mid X \in T_xM \,\}.$$

Proof. Write X as $X \otimes 1 \in T_xM \otimes_{\mathbb{R}} \mathbb{C}$. Then

$$J_x^{\mathbb{C}}\big(X - iJX\big) = J_x^{\mathbb{C}}(X\otimes 1) - i\,J_x^{\mathbb{C}}(JX\otimes 1) = JX\otimes 1 - i\,J(JX)\otimes 1 = JX\otimes 1 + i\,X\otimes 1.$$

On the other hand,

$$i\big(X - iJX\big) = i(X\otimes 1) \;-\; i^2(JX\otimes 1) = X\otimes i \;+\; JX\otimes 1.$$

Since $X \otimes i = i(X \otimes 1)$, the two expressions agree, so

$$J_x^{\mathbb{C}}\big(X - iJX\big) = i\big(X - iJX\big),$$

which shows $X - iJX \in \ker(J_x^{\mathbb{C}} - i\,\mathrm{Id}) = T_x^{1,0}M$. The argument for $X + iJX$ is identical, giving eigenvalue $-i$, hence $X + iJX \in T_x^{0,1}M$. □

We now use the splitting

$$T_x^{\mathbb{C}}M = T_x^{1,0}M \oplus T_x^{0,1}M$$

to introduce (p,q)-type objects. We write $\Lambda^p V$ for the p-th exterior power of a (real or complex) vector space V.

Definition 1.2. For $p, q \geq 0$, the (p,q)-tangent space at $x \in M$ is

$$T_x^{p,q}M \;:=\; \Lambda^p\big(T_x^{1,0}M\big) \;\otimes_{\mathbb{C}}\; \Lambda^q\big(T_x^{0,1}M\big).$$

The (p,q)-tangent bundle is

$$T^{p,q}M \;:=\; \bigsqcup_{x\in M} T_x^{p,q}M.$$

Dually, let

$$T_x^{*\,1,0}M := \big(T_x^{1,0}M\big)^*, \qquad T_x^{*\,0,1}M := \big(T_x^{0,1}M\big)^*.$$

Definition 1.3. For $p, q \geq 0$, the (p,q)-cotangent space at x is

$$\Lambda_x^{p,q}T^*M \;:=\; \Lambda^p\big(T_x^{*\,1,0}M\big) \;\otimes_{\mathbb{C}}\; \Lambda^q\big(T_x^{*\,0,1}M\big).$$

The (p,q)-cotangent bundle is

$$\Lambda^{p,q}T^*M \;:=\; \bigsqcup_{x\in M} \Lambda_x^{p,q}T^*M.$$

A smooth section of $\Lambda^{p,q}T^*M$ is called a (p,q)-form.

Example 1.4. Take $M = \mathbb{C} \cong \mathbb{R}^2$ with its standard almost complex structure J given by

$$J\Big(\frac{\partial}{\partial x}\Big) = \frac{\partial}{\partial y}, \qquad J\Big(\frac{\partial}{\partial y}\Big) = -\frac{\partial}{\partial x}.$$

Then the complexified cotangent space at a point is spanned (over $\mathbb{C}$) by dx and dy.

Define the 1-forms

$$dz := dx + i\,dy, \qquad d\bar{z} := dx - i\,dy.$$

One checks that dz vanishes on $T^{0,1}$ and $d\bar{z}$ vanishes on $T^{1,0}$. Equivalently,

$$dz \in T^{*\,1,0}\mathbb{C}, \qquad d\bar{z} \in T^{*\,0,1}\mathbb{C}.$$

Thus in this model,

$$\Lambda^{1,0}T^*\mathbb{C} = \mathrm{span}_{\mathbb{C}}\{dz\}, \qquad \Lambda^{0,1}T^*\mathbb{C} = \mathrm{span}_{\mathbb{C}}\{d\bar{z}\},$$

which matches our general definition of (p, q)-forms.

In analogy with de Rham theory, we define the exterior derivative

$$d : \Omega^{p,q}(M) \longrightarrow \Omega^{p+q+1}(M),$$

where $\Omega^{p,q}(M)$ denotes the space of smooth (p, q)-forms on M , and $\Omega^{p+q+1}(M)$ here just means forms of total degree $p + q + 1$.

The operator d is $\mathbb{R}$-linear, satisfies $d^2 = 0$, and is determined locally by the usual rules

$$d(f) = \sum_i \frac{\partial f}{\partial x^i}\,dx^i \quad \text{for } f \in C^\infty(M), \qquad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha}\,\alpha \wedge d\beta,$$

where $\deg \alpha = p + q$ if $\alpha \in \Omega^{p,q}(M)$.

We record, without further computations, the following facts.

Proposition 1.5 (Type decomposition of d)**.** *Relative to J, the exterior derivative admits a decomposition*

$$d = \partial + \bar{\partial} + \mu + \bar{\mu},$$

where

$$\partial : \Omega^{p,q} \to \Omega^{p+1,q}, \qquad \bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1}, \qquad \mu : \Omega^{p,q} \to \Omega^{p+2,q-1}, \qquad \bar{\mu} : \Omega^{p,q} \to \Omega^{p-1,q+2}.$$

This decomposition of d looks quite complicated and introduces the extra terms μ and $\bar{\mu}$, which at first sight are not very friendly. In the next section we will see that, although this is the general behavior on an almost complex manifold, there is an important special class, called *complex manifolds*, for which the picture simplifies drastically: in that case one in fact has

$$\mu = \bar{\mu} = 0 \quad \text{and} \quad d = \partial + \bar{\partial}.$$

2. Complex manifolds and the Nijenhuis tensor

We now describe the special class of almost complex manifolds for which the extra terms $\mu, \bar{\mu}$ in the decomposition of d disappear.

Definition 2.1 (Complex manifold)**.** An n-dimensional *complex manifold* is a topological manifold M equipped with an atlas

$$\{(U_\alpha, z_\alpha)\}$$

such that:

- each $U_\alpha \subset M$ is open,
- each $z_\alpha = (z_\alpha^1, \dots, z_\alpha^n) : U_\alpha \to \mathbb{C}^n$ is a homeomorphism onto an open set in $\mathbb{C}^n$,
- whenever $U_\alpha \cap U_\beta \neq \varnothing$, the transition map

$$z_\beta \circ z_\alpha^{-1} : z_\alpha(U_\alpha \cap U_\beta) \longrightarrow z_\beta(U_\alpha \cap U_\beta)$$

  is holomorphic (i.e. each component is holomorphic in the standard sense on $\mathbb{C}^n$).

In such a chart (U_α, z_α), write $z_\alpha^k = x_\alpha^k + iy_\alpha^k$. Then on U_α we obtain a bundle map $J : TM \to TM$ by declaring

$$J\Big(\frac{\partial}{\partial x_\alpha^k}\Big) = \frac{\partial}{\partial y_\alpha^k}, \qquad J\Big(\frac{\partial}{\partial y_\alpha^k}\Big) = -\frac{\partial}{\partial x_\alpha^k},$$

and one checks that this definition is independent of the chosen holomorphic chart. Thus a complex manifold carries a canonical almost complex structure J with $J^2 = -\mathrm{Id}$. We call this J *integrable.*(This is not a standard way of defining this! But we will mention this in the main theorem)

Later we will show that, conversely, an almost complex structure J is integrable (in the sense that it comes from such holomorphic coordinates) if and only if a certain tensor built from J vanishes. This tensor is the Nijenhuis tensor.

Definition 2.2 (Nijenhuis tensor)**.** Let (M, J) be an almost complex manifold. The *Nijenhuis tensor* of J is the $(1,2)$-tensor

$$N_J \ : \ TM \times TM \longrightarrow TM$$

defined on vector fields X, Y by

$$N_J(X,Y) := [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y],$$

where $[\cdot, \cdot]$ is the Lie bracket of vector fields.

Somehow we could just think the tensor N_J measures the failure of J to come from honest holomorphic coordinates.

We now state a fundamental theorem that explains when an almost complex structure actually comes from honest holomorphic coordinates. We will also outline the proof, highlighting the essential ideas.

Theorem 2.3 (Newlander–Nirenberg, equivalent characterizations of integrability)**.** *Let (M, J) be an almost complex manifold. The following are equivalent:*

(1) *The Nijenhuis tensor vanishes:*

$$N_J(X,Y) = 0 \quad \textit{for all vector fields } X, Y.$$

(2) *The exterior derivative has no "torsion terms", i.e.*

$$d = \partial + \bar\partial, \qquad \textit{equivalently } \mu = \bar\mu = 0.$$

(3) *The $(1,0)$-bundle is involutive:*

$$[\Gamma(T^{1,0}M),\, \Gamma(T^{1,0}M)] \subset \Gamma(T^{1,0}M),$$

i.e. $T^{1,0}M$ is closed under Lie brackets of its sections. (Equivalently, $T^{0,1}M$ is involutive.)

(4) *For every local $(1,0)$-form $\alpha \in \Omega^{1,0}(M)$, its exterior derivative has no $(0,2)$-part:*

$$(d\alpha)^{0,2} = 0.$$

(5) *Around every point of M there exist local complex coordinates $(z^1, \dots, z^n)$ such that J acts as multiplication by i in these coordinates. In particular, (M, J) is locally a complex manifold in the sense of holomorphic charts.*

We refer to a J satisfying any (hence all) of the above as *integrable*.

Proof. We indicate how the implications fit together.

(1) $\Leftrightarrow$ (3). Write $X^{1,0} := X - iJX$ and $Y^{1,0} := Y - iJY$. A standard computation shows

$$[JX, JY] - J[JX, Y] - J[X, JY] - [X, Y] \;=\; 4\left([X^{1,0}, Y^{1,0}]\right)^{0,1}.$$

Thus $N_J \equiv 0$ if and only if $[X^{1,0}, Y^{1,0}]$ has no $(0,1)$-part, i.e. lies in $T^{1,0}M$. This is exactly the involutivity condition in (3). This is the Frobenius-type statement: "no $(0,1)$-component of the bracket" $\iff$ "the $(1,0)$-distribution is closed."

(3) $\Rightarrow$ (5). By Frobenius, an involutive complex subbundle such as $T^{0,1}M$ is tangent to a foliation by complex submanifolds. One shows that these leaves give local coordinates in which $T^{0,1}M$ is spanned by $\{\partial/\partial\bar{z}^k\}$ for some complex coordinate functions $z^1, \dots, z^n$. In those coordinates, J acts as multiplication by i on $\partial/\partial x^k$ and $\partial/\partial y^k$. This produces holomorphic charts, which proves (5).

(5) $\Rightarrow$ (2). In holomorphic coordinates $z^k = x^k + iy^k$, the decomposition of forms matches the usual complex notation, and d splits as

$$d = \partial + \bar{\partial} \quad \text{with} \quad \mu = \bar{\mu} = 0.$$

So if such coordinates exist locally around every point, we get (2).

(2) $\Rightarrow$ (4). If $d = \partial + \bar{\partial}$ with no $\mu, \bar{\mu}$ terms, then for any $\alpha \in \Omega^{1,0}(M)$, we have

$$d\alpha = \partial\alpha + \bar{\partial}\alpha \in \Omega^{2,0}(M) \oplus \Omega^{1,1}(M),$$

so there is no $(0,2)$ component. This gives $(d\alpha)^{0,2} = 0$, i.e. (4).

(4) $\Rightarrow$ (1). Take α to be a local $(1,0)$-form that annihilates $T^{0,1}M$ (i.e. $\alpha|_{T^{0,1}} = 0$). Expanding $d\alpha(X^{0,1}, Y^{0,1})$ and using the Cartan formula

$$d\alpha(U, V) = U(\alpha(V)) - V(\alpha(U)) - \alpha([U, V]),$$

one sees that the $(0,2)$-part of $d\alpha$ measures exactly the $(0,1)$-component of $[X^{0,1}, Y^{0,1}]$. Thus $(d\alpha)^{0,2} = 0$ for all $(1,0)$-forms α forces $[X^{0,1}, Y^{0,1}]$ to lie in $T^{0,1}M$, i.e. involutivity (3), hence $N_J = 0$ by the first step. So (4) implies (1).

Combining these implications, we obtain the full chain of equivalences

$$(1) \iff (3) \implies (5) \implies (2) \implies (4) \implies (1),$$

which proves the theorem. □

As a consequence, on an *integrable* almost complex structure, the "bad" terms $\mu, \bar{\mu}$ vanish, d simplifies to $d = \partial + \bar{\partial}$, and the manifold is (locally) a genuine complex manifold.

Remark 2.4. In the proof above, the most subtle step is $(3) \Rightarrow (5)$. There one applies the Frobenius theorem to the involutive subbundle $T^{0,1}M$ in order to produce local complex coordinates. We will not give a full detailed proof of this step here. The other implications are, morally, formal rewritings.

Geometrically this is reasonable: the hardest part is to show that the almost complex structure actually comes from honest holomorphic coordinate charts. Once such coordinates exist, all the algebraic conditions fall into place.

3. Applications of the Nijenhuis tensor

The Nijenhuis tensor N_J may look algebraic and not obviously geometric. However, Theorem 2.3 shows that it is in fact extremely powerful:

$$N_J \equiv 0 \quad \Longleftrightarrow \quad J \text{ is integrable.}$$

This means that, although N_J is defined purely in terms of Lie brackets and J, it can detect the existence of honest holomorphic coordinates. We now give two basic applications.

3.1. **All almost complex structures in real dimension 2 are integrable.** In real dimension 2 the situation simplifies dramatically. In fact, every almost complex structure in dimension 2 is automatically integrable.

Proposition 3.1. *Let (M, J) be an almost complex manifold of real dimension 2. Then $N_J \equiv 0$. In particular, J is integrable.*

Proof. Fix $p \in M$. Choose a local vector field V defined near p such that $V(p) \neq 0$. We first claim that $\{V, JV\}$ is a local frame for TM near p. Indeed, if JV were in the span of V, say $JV = kV$, then applying J again gives

$$-V \;=\; J^2V \;=\; J(JV) \;=\; J(kV) \;=\; k\,JV \;=\; k^2V,$$

which is impossible. So V and JV are linearly independent, hence form a basis.

Now evaluate the Nijenhuis tensor on this basis. By definition,

$$N_J(V, JV) = [V, JV] + J[JV, JV] + J[V, J^2V] - [JV, J^2V].$$

Since $[X, X] = 0$ for any vector field X, and $J^2 = -\mathrm{Id}$, we obtain

$$N_J(V, JV) = [V, JV] + J[V, -V] - [JV, -V] = [V, JV] - J[V, V] + [JV, V] = [V, JV] - [V, JV] = 0.$$

Thus $N_J(V, JV) = 0$ at p. As p was arbitrary, $N_J \equiv 0$ on all of M.

By Theorem 2.3, vanishing of N_J is equivalent to integrability of J, so (M, J) is locally a complex manifold. □

Remark 3.2. Another way to phrase this is: in real dimension 2, the complex line bundle $T^{1,0}M$ has complex rank 1, so any two local $(1,0)$ vector fields are of the form fZ and gZ. Their bracket $[fZ, gZ]$ is again a multiple of Z, hence still of type $(1,0)$. Thus $T^{1,0}M$ is involutive, which by Theorem 2.3 forces $N_J = 0$. This is the Frobenius viewpoint.

3.2. **Holomorphic S^1-actions and complexification.** We now give a second application of N_J. The message is: on a Kähler manifold, a holomorphic S^1-action can be "complexified" to a holomorphic $\mathbb{C}^*$-action. The Nijenhuis tensor is what makes this work cleanly.

Let (M, ω, J, g) be a Kähler manifold, so in particular J is integrable and $N_J = 0$. Suppose we have an S^1-action

$$\rho : S^1 \times M \longrightarrow M$$

which preserves J (and hence the whole Kähler structure):

$$\rho_{e^{i\theta}}^* J = J \quad \text{for all } e^{i\theta} \in S^1.$$

Let X be the fundamental vector field of this action,

$$X_p = \left.\frac{d}{dt}\right|_{t=0} \rho_{e^{it}}(p).$$

Then $L_X J = 0$, i.e. X is real holomorphic.

Because $N_J = 0$, we have the identity

$$(L_{JX}J)(Y) = -(L_X J)(JY) \quad \text{for all vector fields } Y,$$

which comes from rewriting $N_J(X, Y) = 0$ in terms of Lie brackets and Lie derivatives. Since $L_X J = 0$ (the action preserves J), this shows

$$L_{JX}J = 0.$$

So JX is also real holomorphic.

Still using $N_J = 0$ and $L_X J = 0$, we compute

$$(L_X J)(X) = [X, JX] - J[X, X] = [X, JX],$$

so $L_X J = 0$ implies

$$[X, JX] = 0.$$

Thus X and JX commute, and both are real holomorphic vector fields.

Let Φ^θ be the (periodic) flow of X; this is exactly the given S^1-action. Let Ψ^t be the flow of JX. Assume JX is complete, so Ψ^t exists for all $t \in \mathbb{R}$. Since $[X, JX] = 0$, the two flows commute:

$$\Phi^\theta \circ \Psi^t = \Psi^t \circ \Phi^\theta,$$

and since $L_X J = L_{JX} J = 0$, each flow is biholomorphic with respect to J.

We therefore obtain a holomorphic action of $\mathbb{R} \times S^1$ on M by

$$(t, e^{i\theta}) \cdot p := \Psi^t(\Phi^\theta(p)).$$

Using the group isomorphism

$$\mathbb{R} \times S^1 \;\cong\; \mathbb{C}^*, \qquad (t, e^{i\theta}) \longmapsto e^{\,t+i\theta},$$

this defines a holomorphic $\mathbb{C}^*$-action on M extending the original S^1-action.

Proposition 3.3. *Let (M, J) be Kähler, let $\rho : S^1 \times M \to M$ be an S^1-action preserving J, and let X be its fundamental vector field. Then $N_J = 0$ and $L_X J = 0$ imply*

$$L_{JX}J = 0, \qquad [X, JX] = 0.$$

If moreover JX is complete, the S^1-action extends to a holomorphic $\mathbb{C}^*$-action on M.

Remark 3.4. In many geometric settings (e.g. compact Hamiltonian Kähler manifolds), JX is automatically complete. In that case, every holomorphic S^1-symmetry automatically complexifies to a holomorphic $\mathbb{C}^*$-symmetry.

Conclusion

This report began with the introduction of $\mathbb{C}$, viewing "multiplication by i" not just as an algebraic rule but as something that can be promoted to a geometric operator on tangent spaces. This naturally led us to the notion of an almost complex structure, and from there to the decomposition of the complexified tangent bundle and the definition of (p, q)-type tensors and forms.

But the real question of interest was not just how to define these objects, but rather *integrability*: in what sense does an almost complex manifold actually come from honest holomorphic coordinates, i.e. become a complex manifold?

To address that, we introduced the Nijenhuis tensor N_J, and discussed the Newlander–Nirenberg theorem. This theorem provides a deep chain of equivalences stating that

$$N_J = 0 \iff J \text{ is integrable (local holomorphic coordinates exist)} \iff d = \partial + \bar{\partial}$$
$$\iff T^{1,0}M \text{ is involutive.}$$

In other words, the seemingly algebraic condition $N_J = 0$ in fact has a completely geometric meaning: it tells us exactly when an almost complex structure actually comes from complex geometry. We sketched the logic behind these equivalences, though we did not carry out all the technical details (in particular, the step using Frobenius to produce local holomorphic coordinates from involutivity).

After that, we looked at two applications to illustrate why, in practice, people often simply say "integrable $\iff$ $N_J = 0$." First, in real dimension 2, any almost complex structure automatically satisfies $N_J = 0$, so any oriented surface is locally a complex curve. Second, in the Kähler setting, the vanishing of N_J also controls symmetry: an S^1-action preserving J can, under mild assumptions, be extended to a holomorphic $\mathbb{C}^*$-action. This shows that N_J is not only a theoretical integrability test; it can be used to produce concrete geometric structure.

Finally, a comment on perspective. This is a report for a complex analysis course, but the emphasis here was deliberately on the "complex as geometry" point of view rather than on analytic estimates or function theory. I did not attempt to fully develop the analytic side; instead I focused on almost complex structures, integrability, and how complex manifolds emerge from differential-geometric data. This is of course incomplete, but it reflects where my curiosity actually lies: understanding how an operator J on the tangent bundle, together with the condition $N_J = 0$, forces the appearance of holomorphic coordinates and complex structure in a geometric sense.

References